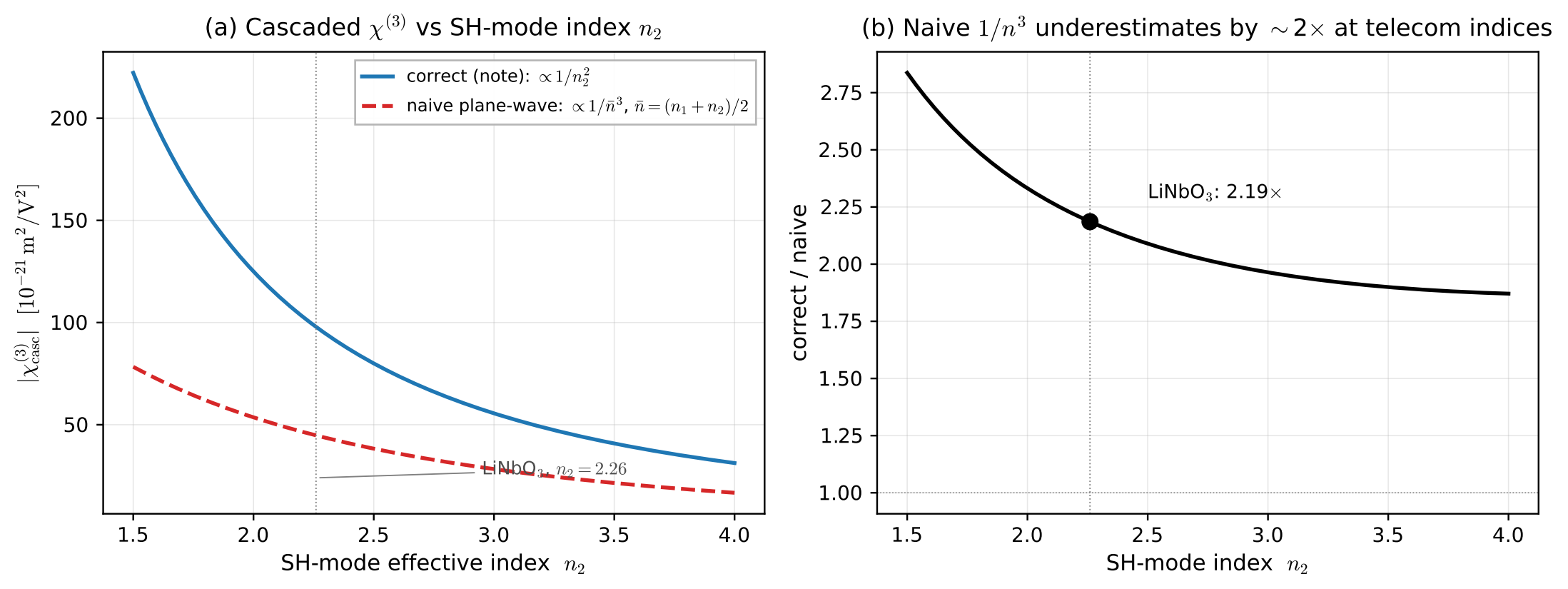
(a) Cascaded $\chi^{(3)}$ vs SH-mode index n_2
correct (note): $\propto 1/n_2^2$
naive plane-wave: $\propto 1/\bar{n}^3$, $\bar{n}=(n_1+n_2)/2$
LiNbO$_3$, $n_2 = 2.26$
$|\chi^{(3)}_{\mathrm{casc}}|$ $[10^{-21}\,\mathrm{m}^2/\mathrm{V}^2]$
SH-mode effective index n_2
(b) Naive $1/n^3$ underestimates by $\sim 2\times$ at telecom indices
LiNbO$_3$: 2.19×
correct / naive
SH-mode index n_2